CAIE Program
CSE382: Introduction to Machine Learning
Spring 2026

Major Task Project: Image Classifiers

Students are encouraged to work in groups. Each group has at most FIVE students.

Due Date: Milestone (1): Week 13 - Milestone (2): Week 15

Major Task Project

Project Overview

Students will design, implement, evaluate, and analyze a complete machine learning pipeline to solve a real-world problem. The project is divided into two progressive phases.

The project emphasizes mathematical understanding, proper experimentation, critical evaluation, and engineering implementation.

Problem Domain

- Image Classification
- Datasets: MNIST.
- Phase 1: Binary Classification (2 Classes)
- Phase 2: Multi classification (10 classes)

Phase 1: Model Development & Baseline System

Objective

Build a complete binary image classification pipeline and implement at least three different machine learning algorithms to distinguish between two selected classes.

Required Tasks

1. Problem Definition

- Define the problem mathematically.
- Define input features and output variable.

2. Data Processing

- Image Resizing and Normalization
- Feature Extraction (Flattening – PCA – HOG)
- Handling class imbalance (if applicable)
- Feature normalization
- Train/validation/test split

3. Model Implementation

Implement at least three algorithms such as:
Logistic/Multinomial Regression

K-Nearest Neighbors (KNN)

Naïve Bayes

Decision Trees
Linear Support Vector Machines
Kernel Support Vector Machines

4. Loss Function & Optimization

• Clearly define the loss function.
• Explain optimization method used.

5. Evaluation
• Classification: Accuracy, Precision, Recall, F1-score, Confusion Matrix

Phase 1 Deliverables

• Clean code
• Technical report (5–7 pages)
• Mathematical formulation
• Experimental results table
• Model comparison discussion

Phase 2: Model Improvement & Intelligent Enhancement

Objective

Scale the pipeline to 10 image classes and enhance baseline performance by applying advanced machine learning techniques.

Required Improvements (Choose at least three)

• Feature engineering: Use pretrained CNN Feature Extraction as input to classical ML classifiers
• Hyperparameter tuning with cross-validation
• Regularization (L1/L2) and bias–variance analysis
• Ensemble methods (Random Forest, Boosting)
• Overfitting/Underfitting diagnosis and learning curves
• Optional: Simple deployment interface

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Final Evaluation Criteria

- Mathematical formulation – 15%
- Implementation correctness – 25%
- Experimental evaluation – 20%
- Improvement strategies – 25%
- Report quality & discussion – 15%

Final Deliverables

- Complete technical report (10–15 pages)
- Clean, commented code
- Presentation (10–15 minutes)
- Demonstration of results

Warnings: (1) Plagiarism is prohibited. (2) Assignments with no reports or no presentations will not be graded.